

INTEGRATION BEE

2026

Official Rulebook

Conducted by **PhySoc**, IIT Kharagpur

1. Introduction

The Integration Bee is a competitive calculus event organized by **PhySoc**, IIT Kharagpur. Participants solve indefinite and definite integrals under time pressure across a series of knockout rounds. The competition runs over **two days**:

Day	Round	Format
Day 1	Round 1 (Screening)	Written, individual
Day 2	Round of 16 (Pre-Quarterfinals)	Head-to-head
Day 2	Round of 8 (Quarterfinals)	Head-to-head
Day 2	Round of 4 (Semifinals)	Head-to-head
Day 2	Final	Head-to-head

2. Schedule and Reporting

Day	Date	Reporting	Start	Notes
Day 1	10 October 2026	6:00 PM	6:30 PM	Round 1. No entry after 6:20 PM.
Day 2	11 October 2026	3:30 PM	4:00 PM	Knockout rounds. Timings of later rounds will be decided on the spot.

- Participants must report at the venue at the **reporting time** on both days.
- On Day 1, **no participant will be allowed to enter after 6:20 PM**, and late arrivals will not be permitted to take part in Round 1.
- On Day 2, the start times of the Round of 8, Semifinals and Final will be announced by the organizers on the spot.

3. Round 1: Written Screening (Day 1)

3.1 Format

- The paper contains **20 questions**.
- The total time allotted is **45 minutes**.
- All participants attempt the paper individually.
- The answer sheet provides **boxes to write the final answer** of each question, along with **space for rough work**.
- There is **no step marking**. Marking is based **only on the final answers** written in the boxes.

3.2 Submission Procedure

1. After finishing, a participant must **raise their hand and call the invigilator**.
2. The invigilator will **immediately collect the paper and record the exact submission time** on it.
3. Once the paper has been collected, **no further changes** are allowed.
4. Participants who do not submit within 45 minutes will have their papers collected at the end of the time limit, with the submission time recorded as 45:00.

3.3 Selection Criteria

1. A total of **16 participants** will qualify for the next round.
2. Participants are ranked by the **number of questions attempted**, with the highest ranked first.
3. **Tie-breaker 1 (Submission time)**: If two or more participants have the same number of questions attempted, the participant with the **lower submission time** (i.e., who submitted earlier) is ranked higher.
4. **Tie-breaker 2 (Degree program)**: If the submission times are also the same, preference is given according to the degree program in the following order:
 - (a) Bachelor's (highest preference)
 - (b) Master's
 - (c) PhD
5. **Tie-breaker 3 (Year of study)**: If the tied participants are in the same degree program, preference is given to the participant in the **lower year** of study.
6. **Tie-breaker 4 (Age)**: If the tied participants are also in the same year, preference is given to the **younger** participant (i.e., the one with the lower age).
7. The final decision of the organizing committee regarding selection is binding.

3.4 Declaration of Results

The results of Round 1 and the list of the 16 qualified participants will be declared **on Day 1 itself, by 11:00 PM**. The selected participants will then compete in the Round of 16 on Day 2.

4. Knockout Rounds (Day 2)

4.1 General Format for Head-to-Head Rounds

1. Participants are divided into **pairs**; each pair competes against each other.
2. All head-to-head matches are conducted on a **blackboard**.
3. Questions are shown **one by one** on a **presentation screen**, each with a timer. The timer is **5 minutes** in the Round of 16 and Round of 8, **6 minutes** in the Semifinals, and **8 minutes** in the Final.
4. Both participants of a pair attempt the displayed question at the same time on the blackboard.

5. **Scoring:** The participant who **completes the question first** gets the score for that question. If **both participants complete the question at the same time**, **both receive the score** for that question.
6. **Winner of a pair:** The participant with the **higher total score** over all the questions shown is the winner of the pair.
7. From each pair, **only one participant** advances to the next round.

4.2 Round of 16 (Pre-Quarterfinals)

1. The 16 qualified participants are divided into **8 pairs**.
2. Each pair solves **5 questions**, with a **5-minute timer** per question.
3. The winner of each pair is promoted to the **Round of 8 (Quarterfinals)**.
4. **Tie-breaker:** If the pair is tied after 5 questions, an **additional question** will be given. This continues until a winner is decided.

4.3 Round of 8 (Quarterfinals)

1. The 8 winners from the Round of 16 are divided into **4 pairs**.
2. The format is **identical to the Round of 16**: 5 questions per pair, **5-minute timer** per question.
3. The winner of each pair advances to the **Semifinals**.
4. **Tie-breaker:** An additional question is given until a winner is decided.

4.4 Round of 4 (Semifinals)

1. The 4 winners from the Quarterfinals are divided into **2 pairs**.
2. The format is the same as earlier rounds, except that each pair solves **7 questions** and the timer is **6 minutes** per question.
3. The winner of each pair advances to the **Final**.
4. **Tie-breaker:** An additional question is given until a winner is decided.

4.5 Final

1. The 2 semifinal winners compete head-to-head.
2. The final consists of **7 questions**, with the same rules, except that the timer is **8 minutes** per question.
3. The participant with the higher total score in the final is declared the **Integration Bee Champion**.
4. **Tie-breaker:** An additional question is given until a winner is decided.

4.6 Summary of Knockout Rounds

Round	Participants	Pairs	Questions	Timer	Tie-breaker
Round of 16	16	8	5	5 min	Additional question
Round of 8	8	4	5	5 min	Additional question
Round of 4	4	2	7	6 min	Additional question
Final	2	1	7	8 min	Additional question

5. General Rules and Conduct

1. Participants must carry **valid ID** and report on time for every round.
2. **Calculators and mobile phones are not allowed** in any round. A participant found with either will be **disqualified**.
3. **Cheating or the use of AI will not be tolerated** and will lead to immediate disqualification.
4. Smartwatches, notes, and books are also prohibited during all rounds.
5. Only the stationery and rough space provided by the organizers may be used, unless stated otherwise.
6. Participants must not communicate with each other during a round.
7. Answers must be **mathematically correct**. Equivalent forms of an answer are accepted. For indefinite integrals, the constant of integration is expected.
8. The judges' decision on the correctness of answers is **final**.
9. Any other form of **misconduct** may also lead to disqualification.
10. Participants must respect the invigilators, judges, and fellow participants at all times.
11. The organizing committee, PhySoc IIT Kharagpur, reserves the right to modify the rules, schedule, or format if required. Any changes will be announced to participants.

Important Reminders

- Round 1: **20 questions, 45 minutes**; top **16** qualify.
- Always call the invigilator on finishing; **submission time is recorded**.
- Knockouts are on the **blackboard**; the first to complete a question scores, and the higher score wins the pair.
- Timers: **5 min** (Round of 16, Round of 8), **6 min** (Semifinals), **8 min** (Final).
- Semifinals and Final: **7 questions**.
- **No calculators, mobiles, or AI**; violations mean disqualification.

Best of luck to all participants!

— Team PhySoc, IIT Kharagpur